Software Requirements Specification for FitFolio: subtitle describing software

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Revision History

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Date 2	1.1	Notes

[All capstone SRS documents should have a reflection appendix. —TPLT]

[This SRS template is based on the IEEE template, with input from co-pilot and S. Smith. It will get you started. You should not modify the section headings, without first discussing the change with the course instructor. Modification means you are not following the template, which loses some of the advantage of a template, especially standardization. —TPLT]

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1 Introduction

This document is the Software Requirements Specification (SRS) for FitFolio, a mobile application in which young adults earn virtual currency from the physical activity their phone records and invest it in a simulated market that follows real-world prices. It states what FitFolio must do and the qualities it must have, without prescribing how it will be built. The SRS is the reference against which the development team and the client agree on scope, the basis for the design, and the source from which the Verification and Validation (V&V) Plan derives its tests. It also records the safety and security requirements identified in the Hazard Analysis (HA), so that design and testing treat them like every other requirement.

The structure of this document follows the course SRS template, which is based on ISO/IEC/IEEE 29148, the international standard for requirements engineering (**?**), and is

consistent with the system life-cycle processes of ISO/IEC/IEEE 15288 (**?**). This section states the purpose and scope of FitFolio, identifies its stakeholders, defines the terms used throughout the document, and describes the knowledge expected of its readers.

1.1 Purpose

FitFolio is intended to help young adults build fitness and financial habits together, at the point in life when starting either one pays off most. It rewards the physical activity users already do with virtual currency, lets them invest that currency in a simulated market at real-world prices, explains financial and fitness concepts where they appear, and lets them share the progress they choose with friends. Because the currency has no real-world value, users can learn by making investing decisions and seeing their results without risking money. The problem that motivates FitFolio is described in detail in the Problem Statement in Section 2.

The same habits also matter beyond the individual user. Canadian guidelines recommend that adults accumulate at least 150 minutes of moderate to vigorous physical activity each week (**?**), and the national financial literacy strategy aims for a Canada in which everyone can build financial resilience (**?**). By making both habits approachable and low-risk, FitFolio is intended to contribute in a small way to both goals.

1.2 Scope

FitFolio is a single product: a smartphone application for Android and iOS, together with the server that supports it.

Problem being solved. Young adults put off building fitness and investing habits because investing advice is confusing and learning by doing risks real money, while beginners at exercise lose motivation before they see results.

Intended users. Adults aged 18 or older, most of them students or people in their first jobs, with little or no investing experience and any level of fitness (Section 2.4).

Major functionality. Within scope, FitFolio:

- reads activity records from the phone's health platform and converts them into virtual currency;
- lets users buy and sell shares of a curated list of real-world stocks and exchange-traded funds (ETFs) at prices that follow the real market;
- shows portfolio value, activity trends, and milestones over time;
- explains financial and fitness concepts where they appear, gives feedback on the user's own decisions, and tracks learning progress through knowledge checks;

- replays past market periods, such as a downturn, as practice scenarios that do not affect the user's live portfolio;
- shares the statistics and milestones that the user selects with accepted friends.

Expected benefits. Users practice investing with no financial risk, stay motivated to exercise because activity has a visible use, and learn core concepts by applying them to their own decisions. The targets that measure these benefits are stated as non-functional requirements.

Out of scope. FitFolio does not:

- handle real money: the currency cannot be bought, withdrawn, or exchanged, and FitFolio never connects to a bank or brokerage account;
- give financial advice: it never recommends buying or selling a real-world security;
- give medical or training advice: it does not diagnose conditions, prescribe exercise programs, or set diet or weight goals;
- integrate directly with wearable devices (a stretch goal), since all activity is read through the phone's health platform;
- offer every listed security, or options, futures, short selling, or borrowing to invest;
- update prices tick by tick: simulated prices follow real prices with a bounded delay, because market data providers limit how often prices can be requested;
- predict future prices: scenarios replay only historical data;
- support public posts, direct messages, or rankings between users;
- run on tablets, the web, or desktop computers.

These limits also bound the inputs FitFolio must handle. For example, a day's step count ranges from zero on a rest day to tens of thousands for a very active user, and every asset price is a positive amount reported by the market data provider.

Standards and regulations. The requirements are written to comply with the following external standards and regulations, which also define the formats of the data FitFolio exchanges:

- **Privacy:** the Personal Information Protection and Electronic Documents Act (PIPEDA) (**?**), which governs the collection, use, and deletion of users' personal and health data.

- **Accessibility:** the Web Content Accessibility Guidelines (WCAG) 2.2 (**?**), applied to the app's screens.
- **Security:** the OWASP Mobile Application Security Verification Standard (MASVS) (**?**).
- **Quality characteristics:** the product quality model of ISO/IEC 25010 (**?**), which names the categories used for the non-functional requirements.
- **Data formats:** JavaScript Object Notation (JSON) (**?**) for data exchanged between the app, the server, and the market data provider; ISO 8601 (**?**) for dates, times, and time zones; and ISO 4217 (**?**) for the currency in which each asset price is quoted.

1.3 Stakeholders

Table 1 lists the stakeholders of FitFolio. Primary stakeholders use the product or define its direction, secondary stakeholders are affected by it indirectly, and external stakeholders are organizations whose services and rules constrain it. Each stakeholder has an identifier, so that every stakeholder requirement in Section 3 can name the stakeholders it comes from.

Table 1: Stakeholders of FitFolio

ID	Stakeholder	Role and interest
Primary		
SH1	End users (beginner investors and fitness-motivated users)	Use FitFolio to earn currency from activity, practice investing, and learn the concepts involved. Want an engaging, low-risk experience that is easy to understand.
SH2	Customer and project sponsor (Chris Schankula)	Proposed the project and defines its vision of gamifying exercise and teaching investing in a low-stakes setting. Approves the scope and validates the product against that vision.
SH3	Development team (Team #1, Burn n' Earn)	Builds and maintains FitFolio. Needs requirements that are complete, unambiguous, and verifiable.
SH4	Administrators	Operate the server after release: manage the list of tradable assets, review educational content, and respond to outages. The development team fills this role; FitFolio has no separate administrator account.
Secondary		
SH5	Friends of users	View the updates a user chooses to share. Expect to see only what the user selected.
SH6	Society	Benefits if more young adults build lasting exercise and financial habits (**??**). Is harmed if the app spreads financial misinformation, encourages unhealthy exercise, or exposes personal data.
External		
SH7	Health platform providers (Apple HealthKit, Google Health Connect)	Supply activity data. Their application programming interfaces (APIs), permission models, and policies limit what FitFolio can read.
SH8	Market data providers	Supply real-world prices. Their accuracy, request limits, and cost limit how closely the simulated market follows the real market.
SH9	App stores and privacy regulators	Set the rules for distributing an app that handles health data and financial content, including PIPEDA (**?**).

1.4 Definitions and Acronyms

Table 2 defines the terms used in this document, in alphabetical order, followed by the acronyms. Each term is used only with the meaning given here. Where an acronym stands for a defined term, its entry links to that definition.

Table 2: Definitions and Acronyms

Term	Definition
Terms	
Activity goal	A daily amount of activity that the user chooses to aim for, such as a number of steps.
Activity record	A measured amount of one activity type over a stated time interval, as reported by the health platform; for example, 4,200 steps between 9:00 and 10:00 on a given day.
Activity type	A category of physical activity recorded by the health platform, such as walking steps, running, cycling, or wheelchair pushes.
Asset	A real-world security that can be traded in FitFolio: either a stock or an exchange-traded fund.
Balance	The amount of virtual currency a user holds and has not invested.
Concept explanation	A short description of a financial or fitness concept, shown where the concept appears in the app.
Diversification	The extent to which a portfolio's value is spread across different assets, so that a fall in one asset has a limited effect on the whole.
Exchange-traded fund	A fund that holds a collection of assets, such as the stocks of many companies, and is itself traded like a stock.
Feedback	An explanation, specific to the user's own portfolio or activity, of what happened and why; for example, why their portfolio's value fell on a given day.
Fitness milestone	A stated level of activity reached by the user, such as their first week with activity recorded every day.
Friend	Another FitFolio user who has accepted a user's friend request.
Health platform	The service built into a smartphone's operating system that stores health and activity data, such as Apple HealthKit on iOS and Google Health Connect on Android.
Holding	The number of shares of one asset that a user owns.
Knowledge check	A short set of questions that tests whether a user understands a concept.
Learning progress	The concepts a user has viewed and the knowledge checks they have passed.
Manually entered record	An activity record that the user typed into the health platform, rather than one measured by a sensor.

Table 2 continued

Term	Definition
Market data provider	An external service that reports current and historical market prices.
Market price	The most recent price of one share of an asset on the real-world exchange where it is listed.
Market scenario	A replay of the price history of a past period, such as a market downturn, in which the user trades with a portfolio separate from their live portfolio.
Portfolio	All of a user's holdings, together with the history of their trades.
Portfolio value	The total value of a user's holdings at their current simulated prices.
Price age	The time elapsed since a simulated price was last updated from the market data provider.
Raw health data	Individual activity records and body measurements (age, height, and weight), as opposed to summaries the user has chosen to share.
Reminder	A notification from FitFolio that prompts the user to be active or to check their portfolio.
Rest day	A day that the user marks as a planned break from activity; it does not end their streak.
Reward	An amount of virtual currency credited to a user for an activity record.
Share	One unit of an asset.
Shared update	A statistic or milestone that a user has chosen to make visible to their friends.
Simulated market	The set of assets that users can trade in FitFolio, together with their simulated prices.
Simulated price	The price at which an asset is traded in FitFolio. In the live market it follows the market price; in a market scenario it follows a historical price.
Social feed	The list of shared updates from a user's friends.
Stock	A share of ownership in a single public company.
Streak	The number of consecutive days on which the user met their activity goal or took a rest day.
Trade	The purchase of shares of an asset for virtual currency, or the sale of shares for virtual currency.
Virtual currency	The unit of value earned in FitFolio. It has no real-world monetary value and cannot be exchanged for money.

Table 2 continued

Term	Definition
Workout	An activity record marked as a deliberate exercise session, with a start time, an end time, and an activity type.
Acronyms	
API	Application Programming Interface
ETF	Exchange-Traded Fund
FR	Functional Requirement
HA	Hazard Analysis
ISO	International Organization for Standardization
JSON	JavaScript Object Notation
MASVS	Mobile Application Security Verification Standard
NFR	Non-Functional Requirement
OS	Operating System
OWASP	Open Worldwide Application Security Project
PIPEDA	Personal Information Protection and Electronic Documents Act
RRSP	Registered Retirement Savings Plan
SH	Stakeholder (identifier prefix in Table 1)
SR	Stakeholder Requirement
SRS	Software Requirements Specification
TFSA	Tax-Free Savings Account
UC	Use Case
V&V	Verification and Validation
WCAG	Web Content Accessibility Guidelines

1.5 Characteristics of Intended Reader

The intended readers of this SRS are the FitFolio development team, the client, course reviewers, and any future maintainer: the people who will review the requirements and design software to meet them. The end users of FitFolio are a different group and are described in Section 2.4. A reader is expected to have:

- **Software engineering:** third-year undergraduate knowledge of software requirements, including functional and non-functional requirements, use cases, traceability, and verification, at the level of a course such as SFWRENG 3RA3 (Software Requirements and Security Considerations).
- **Mathematics:** high-school algebra and percentages, enough to follow how rewards and portfolio values are calculated.

- **Mobile applications:** everyday familiarity with smartphone applications, including installing an app and responding to permission prompts. No mobile development experience is needed.
- **Finance and fitness:** no prior knowledge. Every financial and fitness term the requirements rely on is defined in Table 2.

Because an SRS is a reference document rather than one read from start to finish, readers new to FitFolio are encouraged to begin with the Problem Statement and Product Vision in Section 2, then the Functional and Non-Functional Requirements, and to consult other sections as needed.

2 System Overview

This section provides general information about the system. It identifies the problem, the vision, and the interfaces between the system and its environment. This section also describes user characteristics and lists the system constraints.

[The purpose of this section is to provide general information about the system so that the specific requirements in the later sections will be easier to understand. The general system description section is designed to be changeable independent of changes to the functional requirements documented in the specific system description. The general system description provides a context for a family of related models. The general description can stay the same, while specific details are changed between family members. —TPLT]

2.1 Problem Statement

Current situation. Building good fitness and financial habits early pays off for decades, because both compound over time. Regular exercise builds on itself, and so does money invested early. Poor habits compound in the same way. Yet many young adults, particularly students and those starting their first jobs, struggle to begin either one.

Problem. Advice on managing money is scattered across many sources, which often contradict each other, and learning to invest by doing is risky because beginners' mistakes cost real money. Starting to exercise is similarly daunting, and many beginners lose motivation before they see results.

Impact. Many people put off both habits until later, when poor habits are already established and harder to change.

Desired improvement. An approachable, low-risk way for beginners to build fitness and financial habits together while staying motivated.

2.2 Product Vision

FitFolio is a mobile app in which users earn virtual currency from their everyday physical activity and invest it in a simulated market that follows real-world prices. It lets beginners

build fitness habits and learn to invest without risking real money. Its goals, which the requirements in this document discuss, are:

G1 (Learning by doing). Users learn core financial and fitness concepts through hands-on, low-risk practice, with each decision paired with feedback explaining what happened and why. *Measured by:* a before-and-after knowledge check on a defined set of core concepts, where at least 70% of usability test participants improve their score.

G2 (A human-centered, all-in-one experience). Users see their fitness progress and financial learning together in one place, so they rarely need to leave the app to understand what they are looking at. *Measured by:* usability testing, where at least 80% of first-time participants complete the core tasks without outside help.

G3 (Motivation through friends, with privacy). Users stay motivated by sharing their progress with friends, while keeping control of their data. *Measured by:* users control who sees each shared statistic, and no raw health data is shared unless the user selects it.

G4 (Risk-free investing practice). Users practice investing in a simulated market based on real-world market data, without risking real money. *Measured by:* users can trade at least 50 real-world stocks at prices within 15 minutes of the real market, and can practice in market scenarios such as a replay of a past market downturn.

G5 (Rewards for everyday activity). Users are rewarded for the activities they already do, as recorded by their phone's health platform. *Measured by:* at least five activity types are supported.

2.3 System Context

Figure 1 shows FitFolio as a single system, with its mobile app, server and database inside the boundary, and the external entities it exchanges information with. Activity sources, such as phone sensors, smartwatches and other tracker apps, record activity in the phone's health platform. They never communicate with FitFolio directly, so they are shown outside the boundary with a dashed outline. FitFolio does not connect to banks, brokerages, personal portfolio trackers, or the phone's location services or contacts. Reminders are generated on the phone as local notifications, so FitFolio uses no push notification service.

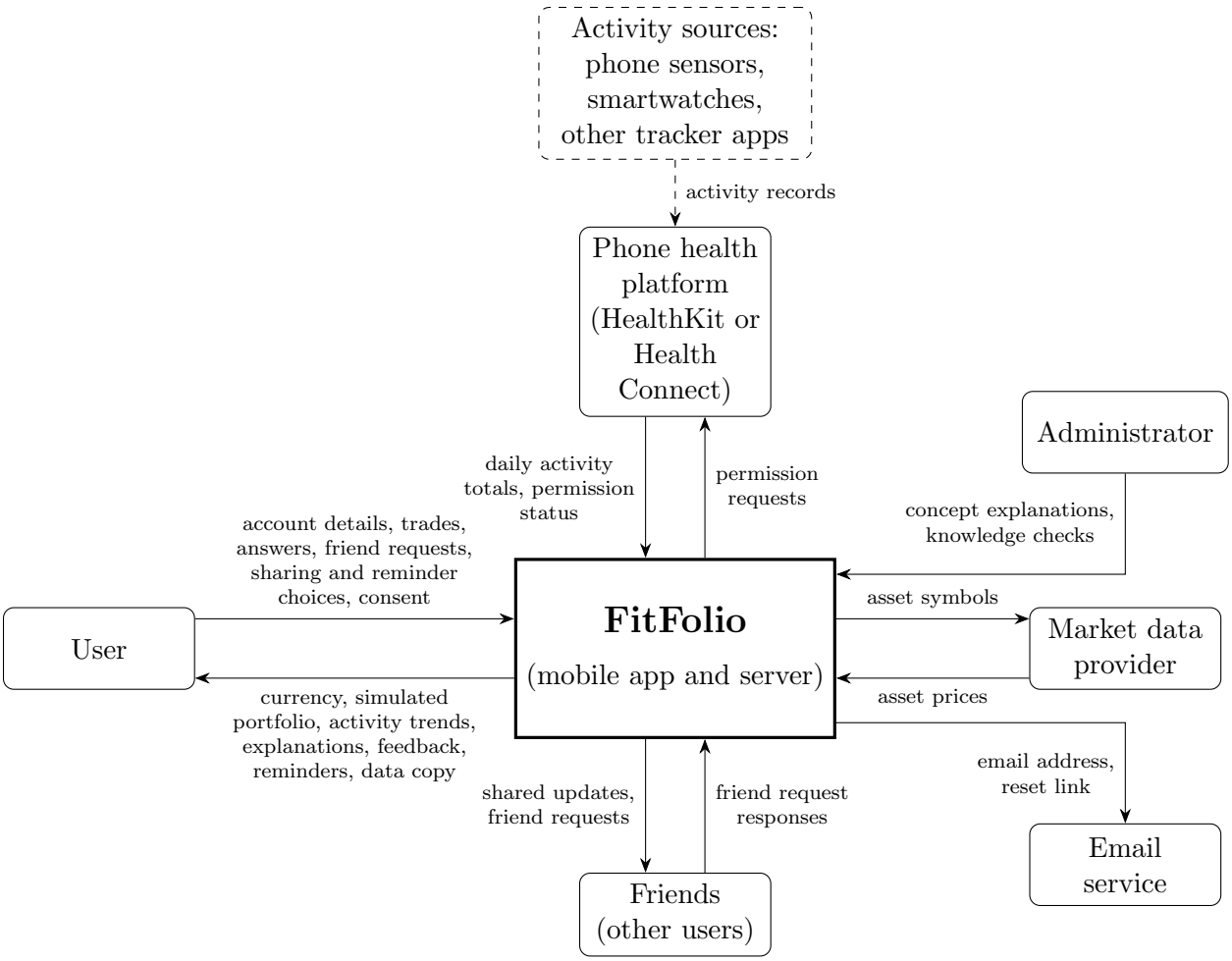


Figure 1: System context of FitFolio

The responsibilities of each entity in Figure 1 are:

- User Responsibilities:
  - Confirm truthfully that they are at least `MIN_USER_AGE` years old.
  - Keep any profile measurements they choose to enter, such as height and weight, accurate.
  - Grant FitFolio the permissions its features use, and keep an activity source recording to the phone's health platform.
  - Make their own trade decisions. FitFolio gives no financial advice.
  - Choose what to share, with which friends, and when to receive reminders.
  - Keep an Internet connection for trading and the social feed.
  - Keep their password private and keep access to their account's email address.

- FitFolio Responsibilities:
  - Reject profile measurements outside plausible ranges.
  - Award currency only for activity within plausibility limits and up to a daily cap (Hazard Analysis SAF-7, SEC-1).
  - Treat a day with no activity data as a day with zero activity.
  - Reject purchases that cost more than the user's balance and sales of more shares than the user holds (Hazard Analysis SEC-5).
  - Record each trade once and completely (Hazard Analysis SEC-5, SEC-6).
  - Show each asset price with the time it was obtained, and mark delayed prices (Hazard Analysis SEC-2).
  - Refuse trades on out-of-date prices while the market is open, and queue trades requested while it is closed (Hazard Analysis SEC-3, SEC-4).
  - Show each shared update only to the friends the user chose, without location detail (Hazard Analysis SEC-13, SEC-14).
  - Request only the permissions its features use, explaining each one (Hazard Analysis SEC-7, SEC-8).
  - Send the market data provider only the symbols of the assets being priced, and no personal or health data (Hazard Analysis SEC-11).
  - Send the email service no personal or health data other than the user's email address and their password reset link (Hazard Analysis SEC-11).
- Health Platform Responsibilities:
  - Store activity records from the user's activity sources, and provide daily totals for the activity types the user has permitted.
  - Grant or withdraw FitFolio's read access as the user chooses, and report the current permission status.
  - Keep the activity data it records as accurate as its activity sources allow.
- Market Data Provider Responsibilities:
  - Supply current and historical prices for the assets FitFolio requests, within its request limits.
- Email Service Responsibilities:
  - Deliver each password reset link that FitFolio sends to the user's email address.
- Administrator Responsibilities:

  - Keep concept explanations and knowledge checks accurate, and name no asset for the user to buy or sell in them (Hazard Analysis SAF-9).

- Friend Responsibilities:

  - Accept or decline a user's friend request.
  - View the updates a user has shared with them in their social feed.

FitFolio is intended for personal education and habit building. Members of the general public use it on their own smartphones to track their activity, earn currency and practice investing in a simulated market. It is not intended for professional, engineering or scientific use.

FitFolio is not mission-critical: it holds and moves no real money, so an outage only delays rewards and learning. It is not safety-critical either since it controls no physical equipment and provides no medical function. However, it handles personal health data and is designed to influence how users exercise and make financial decisions. These hazards, and the requirements that control them, are covered in the Hazard Analysis. The rationale in Section 11 therefore focuses on those requirements.

2.4 User Characteristics

The typical user of FitFolio is a young adult, such as a student or someone starting their first job (Problem Statement and Goals, Section 1). The user of FitFolio:

- Is at least `MIN_USER_AGE` years old (Hazard Analysis CA2).
- Uses a smartphone daily, and can install an app, grant or deny app permissions, and use common mobile app controls.
- Carries an Android or iOS phone during everyday activity. No wearable device or fitness expertise is required.
- Reads English, and has the numeracy of Grade 9 mathematics: working with percentages, comparing values, and reading a simple line chart.
- Needs no prior knowledge of investing terms such as stock, exchange-traded fund (ETF) or portfolio.

As per interviews conducted, participants mainly track their activity with a phone. Both already knew some investing concepts, through self-study or formal study, and both learned most by doing. No one interviewed was a self-reported complete beginner, so the beginner characteristics above come from the Problem Statement, and usability testing should include beginners.

2.5 System Constraints

The following constraints limit the design of FitFolio. Each one is imposed by the project brief, an external service, the law or the app stores.

SC1: FitFolio shall run as a mobile app on Android phones running Android `MIN_ANDROID_VERSION` or later and on iPhones running iOS `MIN_IOS_VERSION` or later. *Rationale:* This project calls for a mobile app, and these are the versions on the devices available to the team and the supervisor for testing (Development Plan, Section 10).

SC2: FitFolio shall obtain recorded activity only through the phone's health platform: Apple HealthKit on iOS and Health Connect on Android. This rule also covers any library FitFolio uses to read health data. No core feature shall require a wearable device. *Rationale:* Users carry their phone during everyday activity (Section 2.4), and the health platforms already collect data from the phone and from any wearable the user owns (note, wearable integration is a stretch goal).

SC3: FitFolio shall access health data only under the access rules Apple (iOS provider) and Google (Android OS provider) set for their health platforms, such as declaring each data type it reads and explaining each permission to the user (**??**). *Rationale:* Apps that break these rules are not accepted in the app stores. The rules also limit what FitFolio can read, such as how far back past activity data can be read.

SC4: FitFolio shall obtain real-world asset prices from a third-party market data provider (`MARKET_DATA_PROVIDER`), within the provider's request limits and licence terms. *Rationale:* The team cannot produce market prices itself, so request limits, cost and permitted use are set by the provider.

SC5: Virtual currency and simulated holdings shall have no monetary value. They cannot be bought with, redeemed for or exchanged for money, and FitFolio shall not connect to any bank or brokerage account or give personalized investment advice. FitFolio shall state the associated risks whenever it explains a financial concept. *Rationale:* Handling real money or giving personalized investment advice would bring FitFolio under financial regulation that the team is not licensed for. Keeping real money out also keeps the practice risk-free (G4).

SC6: FitFolio shall collect, use, store and delete personal and health data in line with Canada's Personal Information Protection and Electronic Documents Act (PIPEDA) and the health-data policies of the Apple App Store and Google Play. *Rationale:* These laws and policies apply to any app that handles this data in Canada and in these stores. The specific controls are in Section 7.3 and the Hazard Analysis (SEC-7 to SEC-12 and SEC-18).

SC7: Trading, market prices and the social feed shall require an Internet connection to FitFolio's server. *Rationale:* Prices are fetched from the provider once for all users to

stay within its request limits (SC4), friends' shared updates must reach other users' phones, and a user's portfolio must survive the loss of their phone.

3 Stakeholder Requirements

This section lists what the stakeholders in Table 1 need from FitFolio, stated without reference to any technology. Each requirement names its *source* stakeholders, the *evidence* for it, and the product *goal* from Section 2 that it supports. Use cases in Section 5 and functional requirements in Section 6 refine these requirements and trace back to them by their identifiers.

The evidence comes from three places. The Problem Statement and Goals document records the client's vision and the target users. Two potential users were interviewed on October 14, 2026, for the team's Design Thinking report; they are referred to as P01 and P02, and their statements are paraphrased or briefly quoted from de-identified transcripts. The Hazard Analysis contributes needs that protect users from harm. Neither interviewee was a beginner at investing, so the needs of beginners come from the Problem Statement and will be checked in usability testing.

Each requirement is verifiable: it can be confirmed by inspection, demonstration, or a usability test, through the functional and non-functional requirements that refine it.

Activity and motivation

SR1: Users shall be rewarded for the physical activity they already do, without having to start, stop, or record it themselves.
Source: SH1, SH2. *Evidence:* Both participants track activity with their phone (P01, P02). Tracking should be "as touchless as possible": an app that must be started and stopped by hand is easily left running for hours and gives wrong results (P01). Approximate step counts are acceptable to users (P01). *Goal:* G5.

SR2: Rewards for activity shall have a visible use to the user beyond the activity count itself.
Source: SH1, SH2. *Evidence:* A long step streak gives one participant nothing but pride (P01); rewards have not changed the other's habits, although a tangible reward might make them work out more (P02). *Goal:* G5.

SR3: Users of any fitness level or physical ability shall be able to earn rewards, whatever kinds of activity they do.
Source: SH1, SH6. *Evidence:* Participants' activity goes well beyond walking: basketball, skating, and pickup hockey (P01); soccer, weightlifting, climbing stairs, and rock climbing (P02). The target users have any level of fitness (Problem Statement). *Goal:* G5.

SR4: Users shall choose their own activity goals, or set none, and shall not be pressured to raise them.

Source: SH1. *Evidence:* One participant has kept the same step goal for over a year, because a higher one takes too much time (P01); the other sets no goals and keeps only a mental check that they "at least did something" (P02). *Goal:* G5.

SR5: Features that motivate activity shall not encourage users to exercise when it is unsafe, such as late at night, while injured, or beyond a healthy amount.
Source: SH1, SH6. *Evidence:* A streak now makes one participant go out walking at 9 p.m. to reach their goal (P01); the other admires a friend for whom "it doesn't matter how much anything hurts" (P02). *Goal:* G5.

Investing practice

SR6: Users shall be able to practise investing without any risk to real money.
Source: SH1, SH2. *Evidence:* One participant was well-versed in financial math from university courses but did not know how to act on it until they started working with real investments (P02); the other first learned by tracking their own transactions by hand (P01). *Goal:* G4.

SR7: Investing practice shall follow the real market closely enough that its lessons carry over to real investing.
Source: SH1, SH2. *Evidence:* The client's vision is a portfolio that follows real-world prices (Problem Statement). *Goal:* G4.

SR8: Users shall be able to practise in past market conditions, such as a downturn, without affecting their own progress.
Source: SH1, SH2. *Evidence:* Practising a downturn safely prepares users for one with real money (Problem Statement). *Goal:* G4.

SR9: FitFolio shall never give, or be mistaken for, advice on what to do with real money.
Source: SH1, SH6, SH9. *Evidence:* Self-taught investors learn from YouTube and Reddit, where it is easy to follow "the wrong kinds of investing YouTubers," and look for answers tailored to their own situation (P01). Users may act on whatever the app appears to recommend. *Goal:* G4.

Learning

SR10: Users shall be able to understand each financial and fitness concept where it appears, without leaving FitFolio.
Source: SH1. *Evidence:* Beginners are told that investing requires studying measures such as price-to-earnings ratios, which confuses them (P01); advice is scattered across sources that contradict each other (Problem Statement). *Goal:* G1, G2.

SR11: Users shall learn through activities they take part in, not only by reading or watching.
Source: SH1. *Evidence:* One participant has trouble learning from videos and needs "a component that's interactive"; watching alone "doesn't stick" (P02). The other learned investing by doing it themselves (P01). *Goal:* G1.

SR12: Users shall be able to find out why their portfolio changed after their own decisions, including how diversified it is.
Source: SH1, SH2. *Evidence:* One participant considers buying individual stocks "almost like gambling" compared with diversified index investing (P01); users need to see this in their own results. *Goal:* G1.

SR13: Users shall be able to learn how the concepts they practise apply to saving and investing in Canada, such as tax-free savings accounts.
Source: SH1, SH6. *Evidence:* Both participants named Canada's account types, such as Tax-Free Savings Accounts (TFSAs) and Registered Retirement Savings Plans (RRSPs), as important things they had to learn (P01, P02); "so many people aren't doing anything at all" with their savings (P02). *Goal:* G1.

SR14: Users shall be able to see how their activity and their investments have changed over time, in one place.
Source: SH1. *Evidence:* One participant checks their heart rate often, out of curiosity and to see that it is healthy (P02); the other tracks both their steps and their portfolio, in separate tools (P01). *Goal:* G2.

Sharing and privacy

SR15: Users shall be able to share the progress they choose with friends.
Source: SH1, SH5. *Evidence:* Being ahead of friends "feels great," and good friends help those who fall behind (P02). *Goal:* G3.

SR16: Users shall control exactly what their friends can see, and nothing shall be shared without the user's choice.
Source: SH1, SH5. *Evidence:* Removing names does not make a person anonymous, because the stories they tell and other context can reveal who they are (P01). *Goal:* G3.

SR17: FitFolio shall not make users feel that they are falling far behind other users.
Source: SH1, SH6. *Evidence:* Comparison is fine when slightly behind, and "only ever bad when you're really far behind" (P02). *Goal:* G3.

SR18: Users' personal and health data shall be collected only where it is needed, kept secure, and never sold.
Source: SH1, SH9. *Evidence:* One participant avoided step counters from large companies, turned off their step counter's network access, and keeps their portfolio records on their own computer (P01); the other shares data only with a reputable company that keeps it secure (P02). Required by PIPEDA (**?**). *Goal:* G3.

SR19: Users shall understand why each permission is needed before they grant it.
Source: SH1, SH7. *Evidence:* Both participants grant a permission when there is a clear reason for it (P01, P02), and distrust apps that ask for "12 permissions" when two would do (P01). *Goal:* G2, G3.

SR20: Users shall be able to withdraw their consent and delete their account and data.
Source: SH1, SH9. *Evidence:* Users' right to withdraw consent is required by PIPEDA (**?**). *Goal:* G3.

Access and operation

SR21: Users shall be able to install FitFolio from their phone's app store and start earning rewards without outside help.
Source: SH1, SH2. *Evidence:* First-time users must complete the core tasks without outside help (Problem Statement Goal 2); how well an app is designed affects whether users trust it (P02). *Goal:* G2.

SR22: Users shall be able to use every core feature with only their phone, on either Android or iOS, without a wearable device.
Source: SH1, SH2. *Evidence:* One participant uses an Android phone and stopped wearing a fitness band because it irritated their skin (P01); the other uses an iPhone and an Apple Watch (P02). *Goal:* G2, G5.

SR23: Users with visual or motor disabilities shall be able to use the core features.
Source: SH1, SH6, SH9. *Evidence:* Accessibility is part of serving every user and is expected by app stores (WCAG 2.2 (**?**)). *Goal:* G2.

SR24: Administrators shall be able to update the tradable assets and the educational content without releasing a new version of the app.
Source: SH4. *Evidence:* Educational content must be corrected and reviewed over time (Hazard Analysis), and the list of assets will change. *Goal:* G1, G4.

SR25: FitFolio shall use health and market data only within the terms and limits set by their providers.
Source: SH7, SH8. *Evidence:* Providers limit what data can be read and how often prices can be requested (Problem Statement). *Goal:* G4, G5.

Table 3 traces each stakeholder requirement to the stakeholders it comes from. Every stakeholder in Table 1 except the development team (SH3), whose needs are expressed through the qualities of the requirements themselves, is the source of at least one requirement.

Table 3: Traceability between stakeholders and stakeholder requirements

	Stakeholder								
Requirement	**SH1**	**SH2**	**SH3**	**SH4**	**SH5**	**SH6**	**SH7**	**SH8**	**SH9**
SR1	×	×							
SR2	×	×							
SR3	×					×			
SR4	×								
SR5	×					×			
SR6	×	×							
SR7	×	×							
SR8	×	×							
SR9	×					×			×
SR10	×								
SR11	×								
SR12	×	×							
SR13	×					×			
SR14	×								
SR15	×				×				
SR16	×				×				
SR17	×					×			
SR18	×								×
SR19	×						×		
SR20	×								×
SR21	×	×							
SR22	×	×							
SR23	×					×			×
SR24				×					
SR25							×	×	

4 User Personas

The personas below represent the main kinds of user FitFolio is built for. They are fictional, but each is based on the evidence named in its *Based on* entry: Personas 1 and 2 on the two interviews (P01 and P02), and Persona 3 on the target users in the Problem Statement, since neither interviewee was a beginner at investing. Details that could identify an interviewee have been changed. Each persona lists the stakeholder requirements that matter most to them, so the design can be checked against the people it serves.

4.1 Persona 1

- **Name:** Sam, 28, graduate student.

- **Description:** Walks to campus and back every day and has tracked steps on an Android phone for years. A streak of more than a year at a fixed step goal now gets Sam out walking at 9 p.m. on slow days, although Sam has never raised the goal. Sam also plays pickup sports and skates in winter, none of which is tracked, and stopped wearing a fitness band because it irritated their skin. Sam taught themselves investing from videos, online forums, and books, and prefers simple, diversified index investing.
- **Goals:** Keep the streak going without it taking over the evening; get credit for sports, not only steps; keep learning investing in a way that fits their own situation; keep control of their data.
- **Frustrations:** Apps that must be started and stopped by hand; investing content that pushes risky stock-picking; apps that ask for far more permissions than they need or that may sell their data.
- **Technical Ability:** High. Sam chooses apps for privacy, turns off an app's network access, and keeps financial records on their own computer.
- **Based on:** P01.
- **Most relevant requirements:** SR1, SR3, SR5, SR9, SR18, SR19, SR22.

4.2 Persona 2

- **Name:** Jordan, 24, early-career employee in financial services.
- **Description:** Took several finance courses at university, but only understood how to act on them once their job involved real investments. Walks, lifts weights now and then, and enjoys rock climbing; used to play soccer. Exercise is not a priority right now. Jordan's iPhone counts their steps and the stairs they climb, and they often check their heart rate on an Apple Watch out of curiosity. Jordan sets no fitness goals and keeps only a mental note of whether they did something that week.
- **Goals:** Be more consistent with exercise, like a friend Jordan admires; learn by doing rather than by watching; understand their account options well enough that their savings grow instead of falling behind inflation.
- **Frustrations:** Learning from videos that do not stick; falling far behind friends; apps that ask for personal data without a clear reason.
- **Technical Ability:** Moderate to high. Uses a phone and smartwatch daily, and judges whether to trust an app by who makes it and how well it is designed.
- **Based on:** P02.
- **Most relevant requirements:** SR2, SR11, SR13, SR15, SR17, SR4, SR19.

4.3 Persona 3

- **Name:** Aisha, 19, first-year student.
- **Description:** Has never invested and does not know what a stock, an exchange-traded fund, or a tax-free savings account is. Her savings sit in a bank account. She walks between classes but does not exercise regularly. She would like to start investing, but is afraid of losing money she cannot afford to lose and has heard that investing means studying company finances.
- **Goals:** Understand the basics of investing in plain language; start a fitness habit without being judged; try things without real consequences.
- **Frustrations:** Financial jargon; advice that contradicts other advice; apps that assume she already knows how they work.
- **Technical Ability:** Everyday smartphone use: she can install an app and respond to permission prompts, but has never used a fitness or investing app.
- **Based on:** the target users described in the Problem Statement, and both interviewees' observations that many people do nothing with their savings (P02) and that beginners are put off by the belief that investing requires expert analysis (P01). This persona will be validated in usability testing.
- **Most relevant requirements:** SR6, SR10, SR13, SR21, SR3, SR14.

5 Use Cases

[Use cases are a way to describe the interactions between users and the system. They are typically written in natural language and describe the steps that a user takes to achieve a specific goal. Use cases can be used to identify functional requirements and to create test cases. —TPLT] [Typically 8-15 substantial use cases. —TPLT]

5.1 UC1: Example Use Case

Primary Actor

TODO

Preconditions

- TODO

Main Success Scenario

1. TODO
2. TODO

3. TODO

Alternative Flows

- TODO

Postconditions

- TODO

6 Functional Requirements

This section lists what FitFolio shall do, grouped by feature area. Each requirement has a fit criterion and traces to the stakeholder requirements and use cases it refines and to the goals in Section 2.2. A requirement that begins with a Hazard Analysis (HA) identifier, such as (HA SEC-3), appears in the Hazard Analysis with the same identifier and wording. A requirement whose traces end with a Hazard Analysis identifier, such as Hazard Analysis SEC-13, implements part of that Hazard Analysis requirement.

6.1 Accounts and Profile

FR1: The system shall let a new user create an account with an email address and a password.
Fit criterion: A test user who enters an unused email address and a valid password has an account, and can sign in with them.
Traces to: SR21, G2

FR2: The system shall not create an account until the user confirms that they are at least `MIN_USER_AGE` years old.
Fit criterion: A test user who does not confirm their age during sign-up has no account created.
Traces to: SR18, G3, Hazard Analysis CA2 and SEC-37

FR3: The system shall require each user to choose a username that no other user has.
Fit criterion: A test user who tries to sign up with a username that another account already uses is asked to choose a different one.
Traces to: SR15, G3

FR4: The system shall let a user sign in with their email address and password.
Fit criterion: A test user who enters their correct email address and password is signed in; one who enters a wrong password is not.
Traces to: SR18, G2

FR5: The system shall let a signed-in user sign out.
Fit criterion: After a test user signs out, opening the app shows the sign-in screen and none of their data.
Traces to: SR18, G3

FR6: The system shall let a user who has forgotten their password set a new one through a link sent to their account's email address.
Fit criterion: A test user who requests a reset receives a link at their email address, and after setting a new password can sign in only with the new one.
Traces to: SR20, G2, Hazard Analysis SEC-40

FR7: The system shall let the user view and edit their profile details.
Fit criterion: A test user who changes a profile detail sees the new value the next time they open their profile.
Traces to: SR20, G3

FR8: The system shall not require the user to enter their height or weight in order to use any feature.
Fit criterion: A test user who leaves height and weight empty can earn currency, trade, learn, and share.
Traces to: SR3, SR18, G3, G5

FR9: The system shall let the user download a copy of the personal data FitFolio stores about them.
Fit criterion: A test user who requests a copy receives a file containing their profile, activity totals, balance, holdings, trades, and shared updates.
Traces to: SR20, G3

FR10: Before deleting an account, the system shall ask the user to confirm the deletion in a message stating that it cannot be undone.
Fit criterion: A test user who starts account deletion is shown this warning, and their account still exists if they cancel.
Traces to: SR20, UC**??**, G3

FR11: The system shall let the user delete their account from within the app.
Fit criterion: A test user who confirms deletion can no longer sign in, and none of their data is shown to them or to their friends.
Traces to: SR20, UC**??**, G3, Hazard Analysis SEC-18

6.2 Activity and Rewards

FR12: The system shall award virtual currency for at least `MIN_ACTIVITY_TYPES` activity types (the supported activity types).
Fit criterion: A tester records activity of `MIN_ACTIVITY_TYPES` different activity types in the health platform, and each one earns currency.
Traces to: SR3, UC**??**, G5

FR13: The system shall award virtual currency for recorded activity of a supported activity type.

Fit criterion: For a test user whose recorded activity in one calendar day is plausible and would earn at most `MAX_DAILY_CURRENCY`, the balance increases by the sum, over the supported activity types, of each type's daily total multiplied by its rate in `EARNING_RATES`.
Traces to: SR1, UC**??**, G5

FR14: (HA SAF-12) Before the user first earns currency, the system shall display a notice stating that FitFolio does not give medical advice and that users should consult a health professional before starting a new exercise program, and shall not award currency until the user acknowledges it.
Fit criterion: A new test user is shown the notice before their first reward, and earns no currency until they acknowledge it.
Traces to: SR5, UC**??**, G5

FR15: (HA SAF-13) The system shall award currency for recorded activity without requiring any interaction with FitFolio while the activity is in progress.
Fit criterion: A test user who records a workout without opening FitFolio during it receives the same currency as one who opens the app during the workout.
Traces to: SR1, UC**??**, G5

FR16: The system shall treat a day with no activity data from the health platform as a day with zero activity.
Fit criterion: For a test user whose health platform reports no activity data for a day, that day shows zero activity and earns no currency.
Traces to: SR1, UC**??**, G5

FR17: (HA SEC-1) The system shall not count an activity record that exceeds the plausibility limits for its activity type toward the user's currency, activity trends, streak or milestones, and shall show the user that the record was not counted.
Fit criterion: A test record of more than `MAX_DAILY_STEPS` steps in one day, or a workout longer than `MAX_WORKOUT_DURATION`, earns no currency, does not appear in the user's activity trends, streak or milestones, and is shown to the user as not counted.
Traces to: SR1, UC**??**, G5

FR18: (HA SAF-7) The system shall award a user no more than `MAX_DAILY_CURRENCY` virtual currency for recorded activity in any calendar day.
Fit criterion: For a test user whose recorded activity in one calendar day is plausible and would otherwise earn more than `MAX_DAILY_CURRENCY`, the balance increases by exactly `MAX_DAILY_CURRENCY` for that day. Each activity counts toward the calendar day in the user's time zone at the time of the activity.
Traces to: SR5, UC**??**, G5

FR19: The system shall let the user set or change a daily activity goal for a supported activity type, with each change applying from the next day.
Fit criterion: A test user who changes their daily activity goal keeps the previous goal

for the rest of that day and has the new goal from the next day.
Traces to: SR4, G5

FR20: (HA SAF-3) The system shall not require the user to set an activity goal in order to earn currency for recorded activity.
Fit criterion: A test user with no activity goal set who records activity receives currency for it.
Traces to: SR4, UC**??**, G5

FR21: (HA SAF-4) The system shall not change a user's activity goal without the user's confirmation.
Fit criterion: A test user who exceeds their goal on `GOAL_REVIEW_PERIOD` consecutive days still has the same goal unless they confirm a change.
Traces to: SR4, G5

FR22: The system shall show the user their current streak.
Fit criterion: A test user with no streak who meets their activity goal on three consecutive days is shown a streak of 3. After a following rest day, they are shown a streak of 4. After a following day on which they neither meet their goal nor take a rest day, they are shown a streak of 0.
Traces to: SR14, G5

FR23: (HA SAF-1) The system shall let the user mark up to `REST_DAYS_PER_PERIOD` days in each `STREAK_PERIOD` as rest days, and a rest day shall not end the user's activity streak.
Fit criterion: A test user with a streak who marks a day as a rest day and records no activity that day does not have the streak reset to zero at the start of the following day. A request to mark more than `REST_DAYS_PER_PERIOD` rest days in one `STREAK_PERIOD` is refused.
Traces to: SR5, G5

FR24: (HA SAF-2) The system shall not reduce a user's virtual currency balance or simulated holdings when they miss an activity goal or their streak ends, and shall not show a missed goal or an ended streak to other users.
Fit criterion: For a test user who misses their goal on day *d*, the currency balance and simulated holdings at the start of day *d+1* equal those at the end of day *d*. No friend's social feed shows the missed goal or the ended streak.
Traces to: SR5, SR17, G3, G5

FR25: The system shall show the user each milestone in `MILESTONES` when they reach it.
Fit criterion: A test user whose streak, or total currency earned, first reaches a value in `MILESTONES` is shown that milestone once.
Traces to: SR14, G5

FR26: (HA SAF-14) The system shall not base any reward, goal or milestone on the user's body weight, change in weight, or calories.
Fit criterion: No reward rule, goal or milestone in FitFolio uses body weight, weight change or calories as an input, and a test user's rewards are unchanged when their weight is changed or removed.
Traces to: SR5, SR3, G5

FR27: The system shall show the user their daily totals for each supported activity type over time.
Fit criterion: A test user with recorded activity on 14 days sees the total for each of those days, for each supported activity type.
Traces to: SR14, G2, G5

6.3 Simulated Market and Portfolio

FR28: The system shall offer at least `MIN_TRADABLE_ASSETS` assets for trading in the simulated market.
Fit criterion: A tester counts the assets that can be traded in the simulated market and finds at least `MIN_TRADABLE_ASSETS`.
Traces to: SR7, UC**??**, G4

FR29: The system shall price each asset in the simulated market at the most recent price it has obtained for that asset from the market data provider.
Fit criterion: For a test asset, the price FitFolio shows and uses for trades equals the latest price FitFolio received from the market data provider for that asset.
Traces to: SR7, UC**??**, G4

FR30: (HA SEC-2) The system shall display each asset price with the time it was obtained, and shall mark the price as delayed when it is older than `QUOTE_STALENESS_LIMIT`.
Fit criterion: A test price obtained more than `QUOTE_STALENESS_LIMIT` ago is shown with its time and a "Delayed" marker. A price obtained within the limit is shown with its time and no marker.
Traces to: SR7, UC**??**, G4

FR31: (HA SAF-8) Before the user's first trade, the system shall display a notice stating that the market is simulated, that virtual currency and holdings have no real-world value, and that nothing in FitFolio is financial advice. The system shall not complete the first trade until the user acknowledges the notice.
Fit criterion: A new test user who attempts their first trade is shown the notice and cannot complete the trade until they acknowledge it. The notice contains all three statements: the market is simulated, currency and holdings have no real-world value, and nothing in FitFolio is financial advice.
Traces to: SR9, UC**??**, G4

FR32: The system shall let the user buy shares of an asset using virtual currency.
Fit criterion: While a test asset's market is open, a test user with enough currency buys one share of it. Their balance decreases by the price shown when they confirmed the trade, and their holding of it increases by one.
Traces to: SR6, SR11, UC**??**, G4

FR33: The system shall let the user sell shares of an asset they hold for virtual currency.
Fit criterion: While a test asset's market is open, a test user holding one share of it sells it. Their balance increases by the price shown when they confirmed the trade, and their holding of it decreases by one.
Traces to: SR6, SR11, G4

FR34: (HA SEC-5) The system shall record each trade completely, updating both the currency balance and the holdings, or not at all. It shall reject a purchase that costs more than the user's currency balance and a sale of more shares than the user holds.
Fit criterion: A test trade interrupted before it completes leaves the balance and holdings unchanged, and a completed trade changes both. A purchase costing more than the balance, or a sale of more shares than are held, is rejected and changes neither.
Traces to: SR7, UC**??**, G4

FR35: (HA SEC-6) The system shall record a trade only once when the same trade request is submitted more than once.
Fit criterion: When a test user submits a trade and the same request is sent again, for example by retrying after a dropped connection, exactly one trade is recorded.
Traces to: SR7, UC**??**, G4

FR36: (HA SEC-3) The system shall not execute a trade while the asset's market is open and the asset's price is older than `QUOTE_STALENESS_LIMIT`.
Fit criterion: While a test asset's market is open and its price is older than `QUOTE_STALENESS_LIMIT`, a trade request for it is refused and the user is told the price is out of date. With a price within the limit, the trade proceeds.
Traces to: SR7, UC**??**, G4

FR37: (HA SEC-4) The system shall queue a trade requested while the asset's market is closed, and execute it at the first price obtained after the market next opens.
Fit criterion: A test trade requested while the market is closed is shown to the user as queued, with a notice that it will run at the opening price, and is executed at the first price obtained after the market opens.
Traces to: SR7, UC**??**, G4

FR38: The system shall show the user each of their holdings.
Fit criterion: A test user who owns shares of two assets sees both holdings, each with the number of shares held.
Traces to: SR14, UC**??**, G2, G4

FR39: The system shall show the user their portfolio value.
Fit criterion: For a test user with known holdings, the portfolio value shown equals the sum of each holding multiplied by its asset's most recent price.
Traces to: SR14, UC**??**, G2, G4

FR40: (HA SAF-10) The system shall label every displayed return and portfolio value as simulated.
Fit criterion: Every screen that shows a return or portfolio value shows a "Simulated" label next to it.
Traces to: SR9, UC**??**, G4

FR41: (HA SAF-18) The system shall show every gain or loss with a sign or a text label, not by colour alone.
Fit criterion: With the display shown in greyscale, a tester can tell every gain from every loss on every screen.
Traces to: SR23, UC**??**, G2, G4

FR42: The system shall let the user trade in a market scenario that replays the historical prices of a past period, such as a market downturn, without changing their own currency balance or holdings.
Fit criterion: A test user starts a scenario that replays a past market downturn, and their trades in it use the historical prices for the replayed dates. After the scenario, their currency balance and holdings outside it are the same as before it started.
Traces to: SR8, SR6, G4

FR43: (HA SAF-15) The system shall mark every screen of a market scenario as a scenario, and shall show the historical dates being replayed.
Fit criterion: Every screen shown during a test scenario carries a scenario label and the dates of the period being replayed, and no screen of the live market does.
Traces to: SR8, G4

FR44: (HA SAF-9) The system shall not recommend that a user buy or sell a specific asset.
Fit criterion: No screen or notification shown to a test user tells them to buy or sell a named asset, or presents assets as a "recommended pick".
Traces to: SR9, UC**??**, G1, G4

6.4 Learning and Feedback

The requirements in this subsection refer to the core concepts in Table 4. These are the concepts measured by the knowledge check in goal G1; the list is educational content that administrators can extend without changing these requirements.

Table 4: Core concepts

Domain	Core concepts
Investing	Stock; ETF; share; market price; portfolio value; return; diversification; risk and volatility; compound growth; market downturn and recovery
Canadian accounts	Tax-Free Savings Account (TFSA); Registered Retirement Savings Plan (RRSP)
Fitness	Each supported activity type; active minutes; rest day; streak

FR45: The system shall provide a concept explanation for each core concept in Table 4.
Fit criterion: For every core concept in Table 4, a tester can open a concept explanation of it in the app.
Traces to: SR10, SR13, G1

FR46: The system shall let the user open the concept explanation of a core concept from every screen on which that concept appears.
Fit criterion: On every screen that shows a core concept, a tester can open its concept explanation without leaving that screen.
Traces to: SR10, G1, G2

FR47: When the user's portfolio value has changed by more than `FEEDBACK_THRESHOLD` since they last viewed their portfolio, the system shall show feedback naming the holdings that contributed most to the change.
Fit criterion: For a test portfolio whose value has changed by more than the threshold, the feedback names the holding whose value changed most.
Traces to: SR12, G1

FR48: After each trade, the system shall show the user how the trade changed the share of their portfolio value held in each asset.
Fit criterion: After a test trade, the feedback shows each holding's share of portfolio value before and after the trade, and these match the values calculated from the holdings.
Traces to: SR11, SR12, G1, G4

FR49: When a single holding is more than `CONCENTRATION_LIMIT` of the user's portfolio value, the system shall show the user an explanation of diversification.
Fit criterion: A test portfolio with one holding above `CONCENTRATION_LIMIT` of its value shows the explanation, and the explanation names no asset to buy or sell (Hazard Analysis SAF-9).
Traces to: SR12, SR9, G1, Hazard Analysis SAF-9

FR50: The system shall provide a knowledge check for each core concept in Table 4.
Fit criterion: For every core concept in Table 4, a tester can start a knowledge check

on it.
Traces to: SR11, G1

FR51: When the user finishes a knowledge check, the system shall show which of their answers were correct and an explanation of the correct answer to each question they got wrong.
Fit criterion: A test user who answers one question wrongly sees that question marked incorrect, together with the explanation of its correct answer.
Traces to: SR12, G1

FR52: The system shall show the user their learning progress.
Fit criterion: After a test user views a concept explanation and passes a knowledge check, both appear in their learning progress.
Traces to: SR14, G1, G2

FR53: The system shall let administrators add and edit concept explanations and knowledge checks without releasing a new version of the app.
Fit criterion: A concept explanation edited by an administrator is shown in its edited form on a test phone whose app has not been updated.
Traces to: SR24, G1

FR54: (HA SAF-16) The system shall show each concept explanation with its source and the date it was last reviewed, and each explanation and knowledge check answer shall be reviewed by a second team member before it is published.
Fit criterion: Every concept explanation shown to a test user has a source and a review date, and the content log shows a second reviewer for every explanation and answer key.
Traces to: SR24, SR10, G1

FR55: (HA SAF-17) The system shall base all feedback on a portfolio change on the user's own holdings and price changes, and shall show the data it is based on.
Fit criterion: For a test portfolio change, the feedback names the holdings and price changes it refers to, and these match the portfolio's records.
Traces to: SR12, G1

6.5 Social Feed

A user can share only the kinds of update in Table 5. Currency balances and portfolio values are not shareable, so that friends compare effort and learning rather than market results.

Table 5: Kinds of shared update

Kind	Content
Activity total	The user's total for one activity type over a day or a week
Streak	The user's current streak, in days. A rest day does not end a streak (Hazard Analysis SAF-1)
Fitness milestone	A fitness milestone the user has reached
Learning milestone	A knowledge check the user has passed

FR56: The system shall let the user send a friend request to another user by their username.
Fit criterion: A test user who enters another user's username sends that user a friend request, which the other user sees.
Traces to: SR15, UC**??**, G3

FR57: The system shall make two users friends only after the user who received the friend request accepts it.
Fit criterion: Two test users are not friends while a request between them is pending or after it is declined, and are friends once it is accepted.
Traces to: SR16, G3, Hazard Analysis SEC-36

FR58: The system shall let the user decline a friend request.
Fit criterion: After a test user declines a request, the two users are not friends and the request is no longer shown to the user who declined it.
Traces to: SR16, G3

FR59: The system shall let the user remove a friend.
Fit criterion: After a test user removes a friend, neither user sees the other's shared updates.
Traces to: SR16, G3, Hazard Analysis SEC-13 and SEC-36

FR60: The system shall let the user share an update of a kind listed in Table 5.
Fit criterion: A test user can share one update of each kind in Table 5, and cannot share anything else.
Traces to: SR15, UC**??**, G3

FR61: The system shall let the user choose which of their friends can see each shared update.
Fit criterion: A test update shared with one selected friend appears in that friend's social feed and in no other friend's.
Traces to: SR16, UC**??**, G3, Hazard Analysis SEC-13

FR62: The system shall share an update only after showing the user the update exactly as their friends will see it.
Fit criterion: For every test update, the preview shown before sharing matches what

the selected friends see in their social feed.
Traces to: SR16, UC**??**, G3

FR63: The system shall not include the user's age, height, or weight in any shared update.
Fit criterion: No shared update shown to a test user's friends contains their age, height, or weight.
Traces to: SR18, G3, Hazard Analysis SEC-35

FR64: The system shall show the user a social feed of the updates their friends have shared with them, newest first.
Fit criterion: A test user's social feed contains every update shared with them by a friend and no others, ordered by the time each was shared.
Traces to: SR15, G3

FR65: The system shall let the user delete an update they have shared.
Fit criterion: After a test user deletes a shared update, it no longer appears in any friend's social feed.
Traces to: SR16, G3, Hazard Analysis SEC-13

FR66: (HA SAF-11) The system shall not rank users against each other by activity, currency or portfolio performance.
Fit criterion: No screen shown to a test user orders users by activity, currency or portfolio performance, or shows a user's position relative to others.
Traces to: SR17, G3

6.6 Reminders and Settings

FitFolio sends reminders of the types in Table 6, which are the same reminder types as in the Hazard Analysis.

Table 6: Reminder types

Type	**Example**	**Feature area**
Activity	Progress toward today's goal, or a streak about to end	Activity and Rewards
Milestone	A goal reached, or a streak or currency milestone	Activity and Rewards
Portfolio	A notable change in simulated portfolio value	Simulated Market and Portfolio
Learning	A new concept explanation or knowledge check is available	Learning and Feedback
Social	A friend shared an update or sent a friend request	Social Feed

FR67: The system shall let the user choose the time periods in which they receive reminders.
Fit criterion: A test user sets a reminder period, and it is shown in their settings the next time they open the app.
Traces to: SR5, UC**??**, G2, Hazard Analysis SAF-6

FR68: (HA SAF-6) The system shall send reminders of the types in Table 6 only during the time periods the user has chosen to receive them.
Fit criterion: For a test user who has chosen their reminder periods, no reminder of any type in Table 6 is delivered outside those periods. For a test user who has not chosen any periods, reminders are delivered only during `DEFAULT_REMINDER_PERIOD`.
Traces to: SR5, UC**??**, G2

FR69: The system shall let the user withdraw consent for FitFolio to read their health data.
Fit criterion: After a test user withdraws consent in the app, no further health data is read.
Traces to: SR20, UC**??**, G3, Hazard Analysis SEC-9

FR70: The system shall show the user which health data types FitFolio has permission to read, and the feature that uses each type.
Fit criterion: A test user who has granted health permissions sees each permitted data type listed with the feature that uses it.
Traces to: SR19, SR18, UC**??**, G3, Hazard Analysis SEC-8

7 Data Requirements

[The Data Requirements section describes what information the system must manage. It includes the major data entities, their attributes, and the relationships between them. It also includes a data dictionary that defines the meaning of each attribute. —TPLT]

7.1 Major Data Entities

[What information must exist for the system to accomplish its goals? —TPLT]
[For each entity, provide a short explanation. —TPLT]

- User
- Song
- Session
- Report

7.2 Data Dictionary

[For each entity, define all important attributes. —TPLT]

Attribute	Type	Description
user_id	UUID	Unique identifier
email	String	User email address

Table 7: Data Dictionary

7.3 Privacy and Data Retention

[Discuss storage, retention, deletion, privacy controls, consent, encryption, and compliance requirements. —TPLT]

TODO

8 Non-Functional Requirements

[You may have to add other NFRs. —TPLT]

[List your nonfunctional requirements. You may consider using a fit criterion to make them verifiable. —TPLT]

[The goal is for the nonfunctional requirements to be unambiguous, abstract and verifiable. This isn't easy to show succinctly, so a good strategy may be to give a "high level" view of the requirement, but allow for the details to be covered in the Verification and Validation document. —TPLT]

[An absolute requirement on a quality of the system is rarely needed. For instance, an accuracy of 0.0101 % is likely fine, even if the requirement is for 0.01 % accuracy. Therefore, the emphasis will often be more on describing now well the quality is achieved, through experimentation, and possibly theory, rather than meeting some bar that was defined a priori. —TPLT]

[You do not need an entry for correctness in your NFRs. The purpose of the SRS is to record the requirements that need to be satisfied for correctness. Any statement of correctness would just be redundant. Rather than discuss correctness, you can characterize how far away from the correct (true) solution you are allowed to be. This is discussed under accuracy. —TPLT]

NFR1: **Accuracy** [Characterize the accuracy by giving the context/use for the software. Maybe something like, "The accuracy of the computed solutions should meet the level needed for <engineering or scientific application>. The level of accuracy achieved by FitFolio shall be described following the procedure given in Section X of the Verification and Validation Plan." If you do this provide a link to the VnV plan. —TPLT]

NFR2: **Usability** [Characterize the usability by giving the context/use for the software. You should likely reference the user characteristics section. The level of usability achieved by the software shall be described following the procedure given in Section X of the Verification and Validation Plan. Include a link to the VnV plan. —TPLT]

NFR3: **Maintainability** [The effort required to make any of the likely changes listed for FitFolio should be less than FRACTION of the original development time. FRACTION is then a symbolic constant that can be defined at the end of the report. —TPLT]

NFR4: **Portability** [This NFR is easier to write than the others. The systems that FitFolio should run on should be listed here. When possible the specific versions of the potential operating environments should be given. To make the NFR verifiable a statement could be made that the tests from a given section of the VnV plan can be successfully run on all of the possible operating environments. —TPLT]

- Other NFRs that might be discussed include verifiability, understandability, reusability, performance, reliability and availability, security, and accessibility. You may have to add other NFRs.

9 AI and Machine Learning Requirements

[Refer to Dr. Mosser's guest lecture on AI requirements. —TPLT]
[Include only when applicable. —TPLT]

9.1 Model Inputs

TODO

9.2 Model Outputs

TODO

9.3 Evaluation Metrics

[Accuracy, Precision, Recall, F1, BLEU, WER, etc. —TPLT]
TODO

9.4 Known Limitations

[Explicitly document expected failure modes. —TPLT]
TODO

10 Requirements Traceability Matrices

[The Requirements Traceability Matrix (RTM) is a table that shows the relationship between different types of requirements. It is used to ensure that all requirements are traceable and that no requirements are missed. The typical flow is: Stakeholder Need -¿ Use Case -¿ Requirement —TPLT]

[There is also a traceability matrix between functional and nonfunctional requirements. The purpose of some FRs is to satisfy NFRs. Therefore, it is important to show the relationship between the two. —TPLT]

[Further traceability to test cases and modules will be given in other documents. —TPLT]

Stakeholder Req	Use Case	Functional Req
SR1	UC1	FR13

Table 8: Traceability Between Stakeholder Requirements, Use Cases, and Functional Requirements

Functional Req	Nonfunctional Req
FR13	NFR1

Table 9: Traceability Between Functional and Nonfunctional Requirements

11 Rationale

[Provide a rationale for the decisions made in the documentation. Rationale should be provided for scope decisions, functional requirements, nonfunctioanl requirements, data requirements, and AI requirements. You don't need a rationale for everything, ust the major decisions that are not obvious. You should have traceability to the entities in the document that you are referencing. —TPLT]

12 Likely Changes

The following changes are identified in the project documents. They describe parts of FitFolio that may change as the team tests its planned integrations or adds features already listed as stretch goals.

LC1: **Activity-data updates**

When FitFolio updates activity data may change because of restrictions imposed by the phone's health platform. The Development Plan identifies updating activity data when

the user opens the app as an alternative if background updates cannot work reliably. This would affect when activity data is retrieved in UC1 (Record Daily Activity) and becomes available for awarding currency under FR13. Any change must still follow the health-platform access rules in SC3.

LC2: **Wearable device support**

FitFolio may be extended to support wearable devices, such as the Apple Watch and other compatible fitness trackers. Wearable integration is already listed as a stretch goal in the Problem Statement and Goals. Adding this support could provide additional activity data, including workout duration and calories burned. This would affect the Activity Record entity in Table **??** and its fields in Table 7. Direct access to a wearable, rather than reading its data through the phone's health platform, would also require revising SC2. Wearables would remain optional, as required by SR22.

LC3: **Market-data access**

The market-data provider, price-update frequency, and selection of supported assets may change if the chosen service's request limits or costs prevent the planned use. The Development Plan identifies switching providers, reducing the update frequency, and limiting the asset list as possible responses. These changes would affect the provider choice in SC4, the prices obtained under FR29, and the available assets represented by the Market Data entity in Table **??**. Changes to the update frequency would also need to account for the price-age display in FR30 and the restriction on trading with stale prices in FR36. Any reduction below the price-freshness or asset-coverage targets in Section 2.2 would require an agreed revision to those targets and the affected requirements.

LC4: **Historical market scenarios**

The number and selection of historical market scenarios may change depending on the availability and cost of historical price data. The Development Plan identifies offering fewer scenarios or using a selected set of stored historical datasets as possible responses if the required data is difficult to obtain. This would affect the past market conditions available under SR8 and the historical prices represented by the Market Data entity in Table **??** and its `price_history` field in Table 7.

LC5: **Social challenges and leaderboards**

FitFolio may be extended to include fitness challenges and opt-in leaderboards. Social challenges and leaderboards are listed as a stretch goal in the Problem Statement and Goals, and the Hazard Analysis identifies opt-in leaderboards as a possible future feature. The current version prohibits ranking users by activity, currency, or portfolio performance under FR66 and Hazard Analysis SAF-11. Adding leaderboards would therefore require an agreed revision to those requirements and a review of the risk of harmful comparison identified in SR17. Adding comparisons of portfolio statistics would also require revisiting the allowed updates in FR60 and Table 5. Users would

still choose whether to participate and which information to share, consistent with SR16 and FR61.

13 Unlikely Changes

The following parts of FitFolio are expected to stay the same because they support the project's purpose and goals.

ULC1: **Virtual currency and simulated investing**

FitFolio is expected to continue using virtual currency with no real-world monetary value. Adding real-money trading or allowing users to exchange earned currency for cash is unlikely. The Problem Statement and Goals describes an app where beginners can practise investing without risking real money or needing to link bank or brokerage accounts. Moving to real money would change this purpose.

ULC2: **Earning rewards through physical activity**

Physical activity is expected to remain the way users earn virtual currency in FitFolio. Removing activity-based rewards is unlikely because earning currency through exercise and using it in the simulated market is the project's central idea. This connection supports the goal of helping users build fitness habits while learning about investing.

ULC3: **User control over social sharing**

Users are expected to keep control over which activity and portfolio information they share. Removing this control is unlikely because the Problem Statement and Goals includes motivating users through friends while protecting their privacy. It also states that raw health data is shared only when the user chooses to share it. Any changes to the social features should preserve this choice.

14 Appendices

14.1 Appendix A: Additional Materials

TODO

14.2 Appendix B: Development Plan

[This section is optional. It is used to explain the plan for developing the software. In particular, this section gives a list of the order in which the requirements will be implemented. In the context of a course this is where you can indicate which requirements will be implemented as part of the course, and which will be "faked" as future work. This section can be organized as a prioritized list of requirements, or it could show the requirements that will be implemented for "phase 1", "phase 2", etc. —TPLT]

14.3 Appendix C: Values of Auxiliary Constants

[Show the values of the symbolic parameters introduced in the report. —TPLT]

[The definition of the requirements will likely call for SYMBOLIC_CONSTANTS. Their values are defined in this section for easy maintenance. —TPLT]

[The value of FRACTION, for the Maintainability NFR would be given here. —TPLT]

Table 10 gives the value of each symbolic constant used in this document and in the Hazard Analysis (HA).

Constant	Value	Used in
MIN_USER_AGE	18	FR2
MIN_ANDROID_VERSION	17	SC1
MIN_IOS_VERSION	27	SC1
MARKET_DATA_PROVIDER	Finnhub (Development Plan)	SC4
MIN_ACTIVITY_TYPES	5 (G5)	FR12
EARNING_RATES	1 per 100 steps. Workouts without a step count, such as cycling or swimming, earn 1 per minute	FR13
MAX_DAILY_CURRENCY	300	FR13, FR18, HA SAF-7
MIN_TRADABLE_ASSETS	50 (G4)	FR28
QUOTE_STALENESS_LIMIT	15 minutes (G4)	FR30, FR36, HA SEC-2, SEC-3
FEEDBACK_THRESHOLD	5%	FR47
CONCENTRATION_LIMIT	50%	FR49

Constant	Value	Used in
DEFAULT_REMINDER_PERIOD	9:00 to 20:00 in the user's time zone	FR68, HA SAF-6
REST_DAYS_PER_PERIOD	2	FR23, HA SAF-1
STREAK_PERIOD	7 days	FR23, HA SAF-1
GOAL_REVIEW_PERIOD	7	FR21, HA SAF-4
MAX_GOAL_INCREASE	10%	HA SAF-5
MILESTONES	Streaks of 7, 30 and 100 days, and 1,000, 10,000 and 100,000 currency earned in total	FR25
MAX_DAILY_STEPS	150,000	FR17, HA SEC-1
MAX_WORKOUT_DURATION	12 hours	FR17, HA SEC-1
SESSION_TIMEOUT	14 days	HA SEC-17
DATA_DELETION_PERIOD	30 days	HA SEC-18
SYNC_LOOKBACK_PERIOD	7 days	HA SEC-20
BACKUP_INTERVAL	24 hours	HA SEC-26
VALUE_TOLERANCE	0.01 currency units	HA SEC-30
MIN_PASSWORD_LENGTH	8 characters	HA SEC-38
MAX_FAILED_SIGNINS	5	HA SEC-38
SIGNIN_LOCKOUT_PERIOD	15 minutes	HA SEC-38

Table 10: Values of auxiliary constants

14.4 Appendix D: Generative AI Disclosure

[Provide full transparency on how generative AI was used in the creation of this document. This includes any text, figures, or tables that were generated by AI. If any skills were created for AI agents to review the documentation, those should be disclosed here. —TPLT]

14.5 Appendix E: Reflection

[This isn't part of the SRS, but it is part of the course. —TPLT]

The information in this section will be used to evaluate the team members on the graduate attribute of Lifelong Learning.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you

to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?
4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.
5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.
6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?

2. What pain points did you experience during this deliverable, and how did you resolve them?

Davis: The main pain point for me was how many design decisions the SRS involved, and how many of them crossed between sections owned by different people. Farid and I split the functional requirements, Angad wrote the use cases and traceability matrices, and Preston wrote the data dictionary, but all of these depend on the same choices, such as what a user can share with friends and whether activity data leaves the phone as individual records or daily totals. Because we each worked in our own branch and only compared our work near the deadline, we found these mismatches late. Some had a ripple effect, for example, we had used different words for the same thing, such as "post" and "shared update", one of the shared tables contradicted a Hazard Analysis requirement on how rest days affect a streak, and no one had filled in the appendix of constants that many of our requirements relied on. As a result, we spent much of the last few days aligning our sections and resolving merge conflicts instead of improving the content. I resolved this by reviewing my teammates' open pull requests against my sections, adopting the terms Farid defined in Section 1.4, filling in the appendix of constants, and listing what I still needed from each person in my pull

request. Next time, I want to communicate with the team earlier and more often. That means agreeing on shared identifiers, terms and the big design decisions at the start of a deliverable, recording each decision in the related GitHub issue as soon as it is made, and merging smaller pieces of work sooner, so that conflicts show up while they are still cheaper to fix.